

Evolutionary Conservation and Phylogenetic Analysis of the Human TP53 Protein Across Mammalian Species

Introduction

The **tumor protein p53 (TP53)** is one of the most extensively studied tumor suppressor proteins in molecular biology and cancer research. Often referred to as the "**guardian of the genome**," TP53 plays a fundamental role in preserving genomic stability by preventing the accumulation and propagation of damaged DNA. The TP53 gene encodes a sequence-specific transcription factor that regulates the expression of numerous genes involved in critical cellular processes, including **cell cycle arrest, DNA repair, apoptosis, senescence, DNA damage response, and cellular metabolism**.

In humans, the **TP53 gene** is located on the **short arm of chromosome 17 (17p13.1)** and encodes a protein consisting of **393 amino acids**. Under normal physiological conditions, p53 is maintained at low intracellular levels through continuous ubiquitination and proteasomal degradation mediated primarily by the E3 ubiquitin ligase **MDM2**. This regulatory mechanism prevents unnecessary activation of p53 in healthy cells.

When cells encounter various forms of stress, such as **DNA damage, oxidative stress, hypoxia, oncogene activation, ribosomal stress, or viral infection**, the interaction between TP53 and MDM2 is disrupted. As a result, p53 becomes stabilized and accumulates within the nucleus, where it binds to specific DNA response elements and activates the transcription of numerous downstream target genes. These genes coordinate cellular responses that either repair damaged DNA or eliminate irreversibly damaged cells through programmed cell death (apoptosis), thereby preventing the transmission of harmful genetic mutations.

Structurally, the human TP53 protein is composed of several highly conserved functional domains, each contributing to its biological activity. The **N-terminal transactivation domain (TAD)** mediates interactions with transcriptional machinery and regulatory proteins. The **proline-rich domain (PRD)** participates in apoptotic signaling and stress responses. The **central DNA-binding domain (DBD)**, spanning approximately amino acid residues **102–292**, is responsible for sequence-specific DNA recognition and represents the most evolutionarily conserved region of the protein. The **oligomerization (tetramerization) domain (OD)** enables p53 to assemble into a functional tetramer, while the **C-terminal regulatory domain (CTD)** modulates DNA-binding activity and is involved in multiple post-translational modifications.

Mutations within the TP53 gene represent the most common genetic alterations observed in human cancers, occurring in approximately **50% of all malignant tumors**. The majority of these mutations are located within the DNA-binding domain and impair the ability of p53 to regulate genes responsible for cell-cycle control and apoptosis. Consequently, defective TP53 function permits genetically unstable cells to proliferate, contributing to tumor initiation, progression, metastasis, and resistance to anticancer therapies. Because of its pivotal role in maintaining cellular homeostasis, TP53 has become one of the most intensively investigated proteins in cancer biology and precision medicine.

Beyond its clinical importance, TP53 exhibits remarkable evolutionary conservation across vertebrate species, particularly among mammals. This high degree of conservation reflects the essential biological functions of the protein and the strong selective pressure to preserve its structural and functional integrity throughout evolution. Comparative analysis of TP53 homologs enables researchers to identify conserved amino acid residues, characterize functionally important protein domains, and investigate evolutionary divergence among species. Such studies provide valuable insights into protein evolution, functional conservation, comparative genomics, and molecular evolution.

In this project, the **human TP53 protein (RefSeq Accession: NP_000537.3)** was selected as the reference sequence for comparative phylogenetic analysis. Homologous TP53 protein sequences from multiple mammalian species were identified using **BLASTp**, aligned using the **ClustalW** algorithm implemented in **MEGA 12**, and analyzed to reconstruct their evolutionary relationships through **Neighbor-Joining phylogenetic analysis**. The resulting phylogenetic tree was subsequently visualized and annotated using the **Interactive Tree of Life (iTOL)** platform to facilitate biological interpretation of TP53 protein conservation across species.

Homolog Identification and Evolutionary Conservation Analysis

Understanding the evolutionary conservation of proteins is fundamental to comparative genomics and molecular evolution. Homologous proteins are sequences that share a common evolutionary origin and often retain similar structural and functional characteristics despite divergence over time. By identifying homologous TP53 proteins from different species, researchers can investigate the extent of sequence conservation, identify functionally important amino acid residues, and reconstruct the evolutionary history of the protein.

In this project, the **human TP53 protein (RefSeq Accession: NP_000537.3)** was selected as the reference (query) sequence for comparative analysis. The TP53 protein was chosen because of its essential role in maintaining genomic stability and its remarkable conservation among vertebrate species. Since proteins evolve more slowly than nucleotide sequences due to the functional constraints imposed on amino acid composition and protein structure, a **protein-based phylogenetic approach** provides a more reliable framework for studying evolutionary relationships across species.

To identify homologous proteins, the reference TP53 protein sequence was retrieved from the **NCBI Protein** database and used as the query sequence in the **BLASTp (Basic Local Alignment Search Tool for Proteins)** program. BLASTp compares an amino acid sequence against a protein database to identify sequences exhibiting statistically significant similarity. Sequence similarity is evaluated using parameters such as **percentage identity**, **query coverage**, **bit score**, and **expectation (E-value)**. Proteins exhibiting high sequence identity, high query coverage, and extremely low E-values are considered strong homologous candidates and are therefore suitable for comparative evolutionary analysis.

The BLASTp search was performed against the **NCBI non-redundant (nr) protein database** using the default search parameters. From the resulting output, the **top 30 homologous TP53 protein sequences** representing diverse mammalian species were selected. Care was taken to include orthologous TP53 proteins while excluding redundant or incomplete protein entries whenever possible. Selecting homologs from multiple taxonomic groups provides broader evolutionary representation and enables the reconstruction of a more informative phylogenetic tree.

The retrieved homologous protein sequences were downloaded in **FASTA** format and combined with the original human TP53 query sequence. This complete dataset served as the input for subsequent **multiple sequence alignment (MSA)** using the **ClustalW** algorithm implemented in **MEGA 12**. Including the query sequence within the alignment allows direct comparison between the reference protein and its homologs, facilitating the identification of conserved amino acid residues, variable regions, and evolutionary divergence among species.

Evolutionarily conserved amino acid residues generally correspond to regions that are critical for protein structure, DNA binding, molecular interactions, or biological function. Conversely, variable regions may represent lineage-specific adaptations or evolutionary divergence that has accumulated over time without substantially affecting the core function of the protein. The identification of these conserved and variable regions forms the basis for subsequent phylogenetic reconstruction and provides valuable insight into the evolutionary history and functional conservation of the TP53 protein across mammalian species.

The homolog identification process represents the foundation of this comparative bioinformatics study. The quality of the selected homologous sequences directly influences the accuracy of the multiple sequence alignment and the reliability of the resulting phylogenetic tree. Therefore, careful sequence selection using biologically meaningful similarity criteria is essential for generating robust evolutionary inferences and understanding the conservation of the TP53 protein.

Methodology

The **human TP53 protein sequence** (RefSeq Accession: **NP_000537.3**) was retrieved from the **NCBI Protein** database and used as the reference sequence for this study. A **BLASTp** search was performed against the NCBI non-redundant (nr) protein database to identify homologous TP53 proteins from diverse mammalian species. The top homologous protein sequences were selected based on sequence similarity, query coverage, and E-value, and downloaded in FASTA format.

The retrieved protein sequences, together with the human TP53 query sequence, were imported into **MEGA 12**, where a **multiple sequence alignment (MSA)** was performed using the **ClustalW** algorithm. The aligned sequences were subsequently used to construct a **Neighbor-Joining (NJ)** phylogenetic tree with **1,000 bootstrap replicates** to assess the reliability of the inferred evolutionary relationships. Finally, the phylogenetic tree was exported in **Newick (.nwk)** format and visualized using the **Interactive Tree of Life (iTOL)** platform for annotation and presentation.

Tools Used

Tool/Database	Purpose
NCBI Protein	Retrieval of the human TP53 protein sequence (NP_000537.3).
BLASTp	Identification of homologous TP53 protein sequences based on sequence similarity.
MEGA 12	Multiple sequence alignment (ClustalW) and phylogenetic tree construction using the Neighbor-Joining method.
iTOL (Interactive Tree of Life)	Visualization, annotation, and customization of the phylogenetic tree for interpretation and presentation.

Figure 1.1: Showing all Tools & Databases used in project.

Homologous Protein Identification and Sequence Retrieval

To investigate the evolutionary conservation of the TP53 protein, the **human TP53 protein sequence** (RefSeq Accession: **NP_000537.3**) was retrieved from the **NCBI Protein** database and used as the reference (query) sequence for homolog identification. The query sequence was submitted to the **BLASTp (Basic Local Alignment Search Tool for Proteins)** program to search against the **NCBI non-redundant (nr) protein database**, enabling the identification of homologous TP53 proteins from a diverse range of mammalian species.

The BLASTp search was performed using the default search parameters. Homologous protein sequences were evaluated based on key statistical measures, including **percentage identity**, **query coverage**, and **E-value**. Protein sequences exhibiting high sequence identity, high query coverage, and statistically significant E-values were considered reliable homologs and selected for further analysis.

From the BLASTp output, the **top 30 homologous TP53 protein sequences** representing different mammalian species were selected to provide a comprehensive dataset for comparative evolutionary analysis. Care was taken to include complete, well-annotated protein sequences while avoiding redundant entries. The selected homologous protein sequences were downloaded in **FASTA** format and combined with the original human TP53 query sequence.

The resulting multi-FASTA dataset served as the input for subsequent **multiple sequence alignment (MSA)** and phylogenetic analysis. Including the human TP53 query sequence alongside its homologs enabled direct comparison of amino acid conservation, identification of functionally important residues, and reconstruction of the evolutionary relationships among TP53 proteins across different mammalian species.

	A	B	C	D	E	F	G	H	I	J	K	L	M
	Query Accession	Subject Accession	% Identity	Alignment Length	Mismatches	Gap Opens	Query Start	Query End	Subject Start	Subject End	E-value	Bit Score	% Positives
2	NP_000537.3	NP_000537.3	100	393	0	0	1	393	1	393	0	813	100
3	NP_000537.3	XP_050621303.1	95.674	393	17	0	1	393	98	490	0	785	97.2
4	NP_000537.3	NP_001119590.1	100	354	0	0	40	393	1	354	0	737	100
5	NP_000537.3	XP_011813950.1	89.822	393	16	1	1	393	1	369	0	724	91.09
6	NP_000537.3	NP_001119585.1	97.654	341	8	0	1	341	1	341	0	689	98.24
7	NP_000537.3	XP_008008383.3	94.565	368	15	1	26	393	44	406	0	689	95.38
8	NP_000537.3	KAL2765772.1	88.55	393	41	2	1	393	1	389	0	687	90.33
9	NP_000537.3	XP_012631512.1	86.005	393	47	3	1	393	1	385	0	664	88.55
10	NP_000537.3	XP_062967318.1	87.786	393	46	1	1	393	1	391	0	664	91.6
11	NP_000537.3	XP_032136321.1	90.541	370	33	1	26	393	43	412	0	662	93.51
12	NP_000537.3	NP_001306217.1	84.518	394	57	2	1	393	1	391	0	661	89.09
13	NP_000537.3	XP_069338125.1	82.952	393	37	3	1	393	1	363	0	654	86.01
14	NP_000537.3	NP_001075873.1	86.041	394	51	3	1	393	1	391	0	651	90.36
15	NP_000537.3	XP_012661495.1	83.969	393	60	2	1	393	1	390	0	647	87.79
16	NP_000537.3	9CHT_8	98.718	312	4	0	1	312	85	396	0	646	99.36
17	NP_000537.3	XP_064234431.1	89.674	368	36	1	1	366	1	368	0	638	92.93
18	NP_000537.3	XP_065753056.1	80.964	394	48	2	1	393	1	368	0	637	84.52
19	NP_000537.3	AYE20624.1	100	305	0	0	1	305	1	305	0	635	100
20	NP_000537.3	XP_052612630.1	82.532	395	63	4	1	393	1	391	0	634	86.33
21	NP_000537.3	AHK05716.1	82.576	396	53	5	1	393	1	383	0	634	86.11
22	NP_000537.3	XUP55605.1	80.303	396	66	6	1	393	1	387	0	632	85.61
23	NP_000537.3	KAL2765776.1	89.548	354	33	2	40	393	1	350	0	630	90.96
24	NP_000537.3	XP_059988823.1	84.264	394	54	3	1	393	63	449	0	629	88.07
25	NP_000537.3	NP_001297199.1	83.206	393	64	1	1	393	1	391	0	628	87.28
26	NP_000537.3	XP_036053163.1	81.864	397	64	5	1	393	112	504	0	625	86.15
27	NP_000537.3	KAL1779023.1	78.481	395	78	5	1	393	1	390	0	623	84.56

Figure 1.2: BLASTp search results showing the top homologous TP53 protein sequences retrieved from the NCBI non-redundant protein database.

Multiple Sequence Alignment of TP53 Homologs using MEGA 12

The homologous **TP53 protein sequences** retrieved from the BLASTp analysis were subjected to **multiple sequence alignment (MSA)** using the **ClustalW** algorithm implemented in **MEGA 12 (Molecular Evolutionary Genetics Analysis)** software. Multiple sequence alignment is a fundamental step in comparative bioinformatics, enabling the identification of conserved and variable amino acid residues among homologous proteins while providing the basis for accurate phylogenetic reconstruction.

The downloaded homologous TP53 protein sequences, together with the **human TP53 protein query sequence (RefSeq Accession: NP_000537.3)**, were imported into the MEGA 12 Alignment Explorer in FASTA format. The ClustalW algorithm was executed using the default alignment parameters to align the amino acid sequences according to their evolutionary similarity. Gap insertion and extension penalties were automatically applied to optimize sequence alignment while preserving homologous positions across all protein sequences.

Following alignment, the resulting multiple sequence alignment was carefully examined to identify **highly conserved**, **moderately conserved**, and **variable** regions within the TP53

protein family. Conserved amino acid residues indicate regions that have remained largely unchanged throughout evolution due to strong functional or structural constraints, whereas variable regions reflect evolutionary divergence and species-specific adaptations. Particular attention was given to the conservation of residues within the DNA-binding domain, as this region is essential for the biological activity of the TP53 protein.

The completed alignment served as the primary dataset for subsequent phylogenetic analysis. Accurate sequence alignment is critical because the reliability of phylogenetic tree construction depends directly on the correct positioning of homologous amino acid residues. The aligned sequences were therefore saved in **MEGA (.meg)** format for downstream evolutionary analysis and phylogenetic tree construction.



Figure 1.3 : Multiple sequence alignment of TP53 homologous protein sequences generated using the **ClustalW** algorithm in **MEGA 12**, illustrating conserved and variable amino acid residues across selected mammalian species.



Figure 1.4: Conserved region of the multiple sequence alignment of TP53 homologous proteins generated using the **ClustalW** algorithm in **MEGA 12**, demonstrating a high degree of amino acid conservation across mammalian species with only minor sequence variations.

Phylogenetic Tree Construction of TP53 Homologs using MEGA 12

Following multiple sequence alignment, the aligned **TP53 protein sequences** were used to reconstruct the evolutionary relationships among the selected mammalian species through **phylogenetic analysis**. The alignment generated in **MEGA 12** served as the input dataset for tree construction, ensuring that homologous amino acid positions were accurately compared across all sequences.

A **Neighbor-Joining (NJ)** phylogenetic tree was constructed using **MEGA 12**, a widely used software package for molecular evolutionary analyses. The Neighbor-Joining algorithm is a distance-based method that estimates evolutionary relationships by clustering sequences according to their genetic distances while minimizing the total branch length of the tree. Evolutionary distances were computed using the **Poisson correction model**, which estimates amino acid substitutions while accounting for multiple substitutions occurring at the same site during evolution.

To evaluate the statistical reliability of the inferred phylogenetic relationships, a **bootstrap analysis with 1,000 replicates** was performed. Bootstrap values were calculated for each internal node of the tree, providing a measure of confidence for the observed evolutionary groupings. Branches supported by higher bootstrap values indicate greater confidence in the corresponding evolutionary relationships among TP53 homologs.

The resulting phylogenetic tree revealed the evolutionary relationships among TP53 proteins from different mammalian species. Closely related species clustered together, reflecting their shared evolutionary ancestry and the high conservation of the TP53 protein. The tree also illustrated the evolutionary divergence among more distantly related mammals while demonstrating the remarkable conservation of this essential tumor suppressor protein across the analyzed species.

The completed phylogenetic tree was saved in **MEGA format** for future analysis and exported in **Newick (.nwk)** format to facilitate downstream visualization and annotation using the **Interactive Tree of Life (iTOL)** platform.

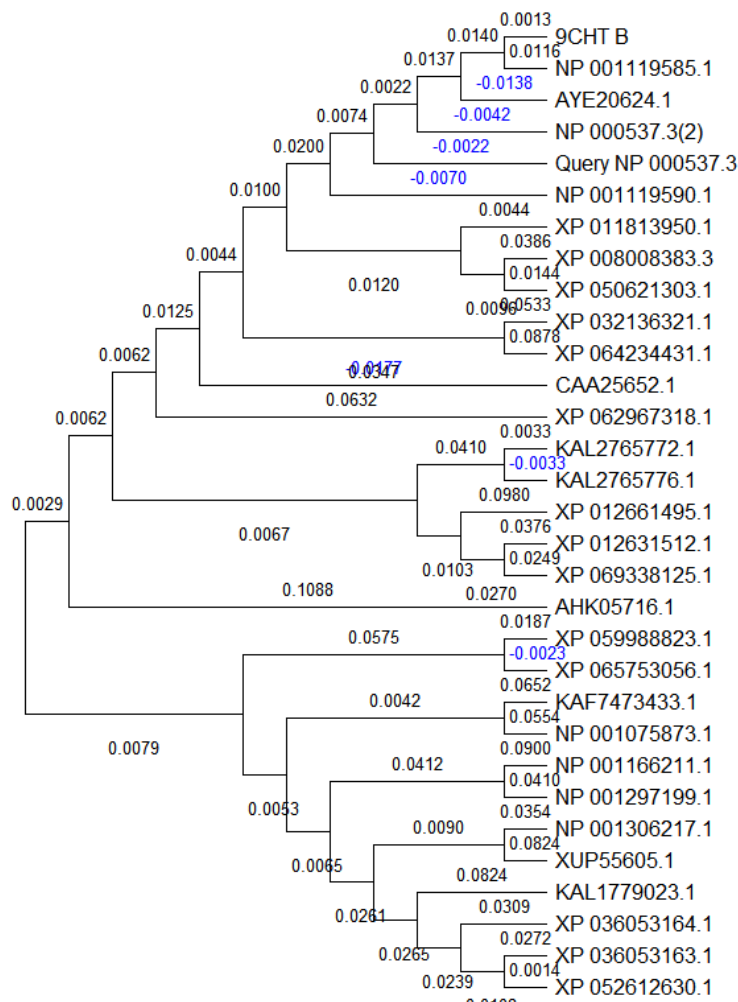


Figure 1.5: Bootstrap-supported phylogenetic tree illustrating the evolutionary relationships among TP53 proteins from different mammalian species.

Phylogenetic Tree Visualization and Annotation of TP53 Homologs using iTOL

Following the construction of the **Neighbor-Joining (NJ) phylogenetic tree** in **MEGA 12**, the resulting tree was exported in **Newick (.nwk)** format for visualization and annotation. The Newick file was then uploaded to the **Interactive Tree of Life (iTOL)** platform, an online tool widely used for the interactive visualization, customization, and annotation of phylogenetic trees.

Within iTOL, the phylogenetic tree was refined to improve its clarity and presentation. Various customization features were applied, including optimization of the tree layout, adjustment of branch lengths, formatting of species labels, and display of bootstrap support values generated during phylogenetic analysis. These modifications enhanced the readability of the tree while preserving the inferred evolutionary relationships among the TP53 homologs.

To facilitate biological interpretation, species names were clearly labeled and branches representing closely related taxa were visually distinguished. A scale bar representing evolutionary distance was included to illustrate the relative genetic divergence among the analyzed TP53 proteins. The annotated tree provided a clear representation of the clustering patterns among mammalian species and highlighted the remarkable evolutionary conservation of the TP53 protein across different evolutionary lineages.

The finalized phylogenetic tree was exported as a **high-resolution image** suitable for publication and project documentation. Visualization using iTOL not only improved the overall presentation of the phylogenetic analysis but also enabled easier interpretation of evolutionary relationships, conserved protein lineages, and species-specific divergence within the TP53 protein family.

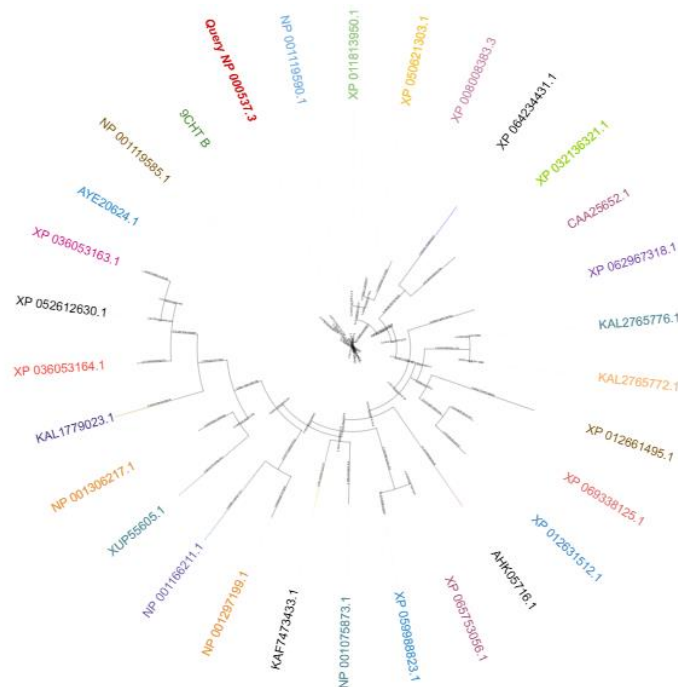


Figure 1.6: Annotated phylogenetic tree of TP53 homologous proteins visualized using **iTOL**, illustrating the evolutionary relationships among selected mammalian species.

Conclusion

This study successfully investigated the evolutionary conservation of the **human TP53 protein** through a comprehensive comparative bioinformatics workflow involving **NCBI Protein**, **BLASTp**, **MEGA 12**, and **Interactive Tree of Life (iTOL)**. The human TP53 protein sequence (RefSeq Accession: **NP_000537.3**) was used as the reference sequence to identify homologous TP53 proteins from diverse mammalian species and reconstruct their evolutionary relationships.

The BLASTp analysis identified highly similar TP53 homologs, with sequence identities ranging from approximately **78% to 100%**, high query coverage, and extremely significant **E-values (0.0)**, demonstrating the strong evolutionary conservation of the TP53 protein across mammals. These findings indicate that TP53 has been subjected to strong evolutionary constraints, reflecting its indispensable role in maintaining genomic integrity and regulating critical cellular processes such as DNA repair, cell-cycle arrest, apoptosis, and cellular senescence.

Multiple sequence alignment performed using **ClustalW** in **MEGA 12** further confirmed the presence of highly conserved amino acid residues among the selected TP53 homologs while also revealing limited sequence variation associated with evolutionary divergence among species. The conserved regions correspond to functionally important domains of the TP53 protein,

emphasizing the importance of preserving its structural and biological functions throughout mammalian evolution.

The **Neighbor-Joining phylogenetic tree** constructed from the aligned protein sequences accurately grouped evolutionarily related mammalian species, demonstrating that phylogenetic relationships inferred from TP53 protein sequences are consistent with established evolutionary classifications. Subsequent visualization and annotation using **iTOL** enhanced the interpretation of the phylogenetic tree by providing a clear and informative representation of species clustering and evolutionary distances.

Overall, this project demonstrates the effectiveness of integrating **protein sequence retrieval, homology searching, multiple sequence alignment, phylogenetic reconstruction, and tree visualization** into a unified comparative bioinformatics workflow. The results confirm that the TP53 protein is highly conserved among mammals, highlighting its essential biological role and evolutionary significance. Furthermore, this study provides valuable practical experience in protein phylogenetics and illustrates how computational approaches can be used to investigate molecular evolution, functional conservation, and evolutionary relationships among homologous proteins.